

Analysing the Sales Trends of Various Food and Beverage Items in CCD, IITM Branch

BDM Capstone Project Mid-Term Submission

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1. Executive Summary

Recap & Business Problem

CCD@IITM faces inventory and revenue optimisation challenges. This project analyses ~90 days of sales data to uncover trends and drive actionable insights. The CCD branch caters to students, faculty, and staff, operating under pricing regulations imposed by the Committee for Monitoring General Facilities for Students (CMGFS), which results in significantly subsidised pricing—up to 50% lower than other outlets. Although this makes CCD an affordable option on campus, it has led to recurring financial constraints and necessitates improvements in operational efficiency, particularly in inventory management.

Data & Scope

I used the CCD billing system dataset (Aug 23–Dec 19 2024; 800 rows, 6 columns). I also gathered information on the inventory restocking patterns.

Key Descriptive Findings

Some noteworthy observations are that the average daily revenue per item is Rs. 172.30. Iced mocha, which is the highest grossing item is sold at 28 units a day and ironically, food items account for 60% of the revenue!

Methods & Insights Overview

We apply descriptive statistics, Pareto, time-series, and correlation analyses. Top 8 items generate 82% of revenue; weekends see 40% uplift; snack–beverage correlation $r=0.10$ suggests combo potential.

2.Proof of Originality

Details:

Café Coffee Day, Near Gajendra Circle

Delhi Avenue, Indian Institute of Technology

Chennai- 600036Location on the map:

[Café Coffee Day](#)

Videos of the Conversation with CCD Incharges of the IITM branch:

[Conversation with the Head of Operations of CCD for Tamil Nadu, Mr. Jagadish](#)

[Conversation with Mr. Harish Jagadish, CCD-IITM manager of five years](#)

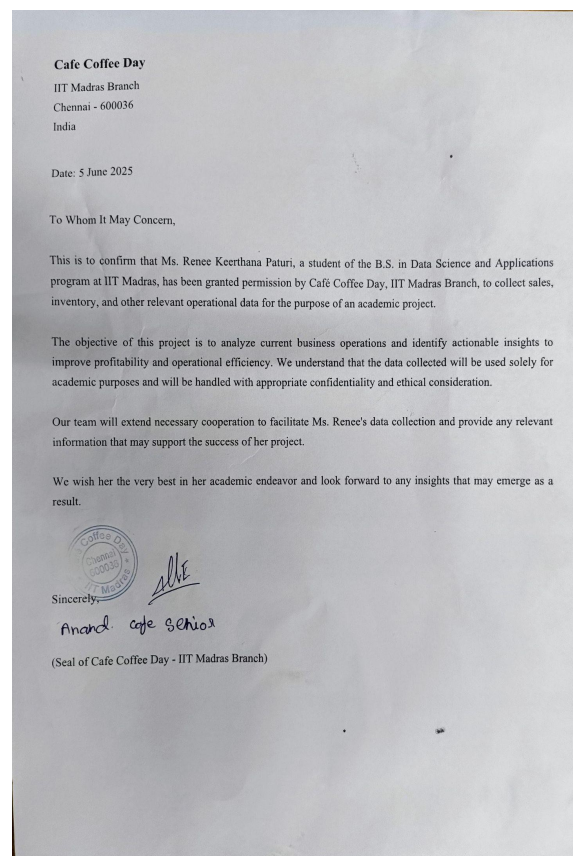


Figure 2.1: Letter of Approval



Figure 2.2: CCD from the Outside



Figure 2.3



Figure 2.4



Figure 2.5: Chocolate Nirvana Sundae

COFFEE DAY GLOBAL LIMITED			
IIT MADRAS INDIAN INSTITUTE OF TECHNOLOGY, MADRAS, IIT P.O., CHENNAI - 600008			
DATE : 2024-08-25 00:15:11			
EMP CODE : a180553			
TOTAL SALES	55542.02		
TOTAL TAX	7777.10		
GROSS SALES	56319.12		
DISCOUNT	0.00		
NET SALES	56319.00		
CASH TO BE DEPOSITED	6985.00		
CASH	6985.00		
Denomination	Qty	Total	
500	13.00	6500.00	
100	4.00	400.00	
1	05.00	05.00	
Total	22.00	6905.00	
	QTY	PRICE	
Sub Total	0	0	
Audit	899.0000	1.54	
Beverages	247.0000	48.39	
Combs	0	0	
Food	298.0000	46.57	
Grab & Go	37.0000	3.50	
Others	0	0	
COFFEE DAY GLOBAL LIMITED IIT MADRAS INDIAN INSTITUTE OF TECHNOLOGY, MADRAS, IIT P.O., CHENNAI - 600008			
LINE : 0013-52 DATE : 25-Aug-2024			
SALES RECEIPT FROM 24-08-2024 TO 24-08-2024			
CUSTOMER / SUBCUSTODY			
ITEM NAME	PRICE	QTY	SALE AMT
AUDIT			
AUDIT			
AUDIT GIFT CARD 0.0524	899	0.00	899.02
AUDIT TOTAL	899	0.00	899.02
AUDIT TOTAL	899	0.00	899.02
BEVERAGES AND SNACKS			
FLAVOROUS / SMOO			
FLAVOROUS / SMOO 25.0000	2	70.00	70.56
FLAVOROUS / SMOO 25.00	2	70.00	70.56
BEVERAGE AND SNACK 25.00	2	70.00	70.56
BIG EATS			
CROISSANTS			
CHOCOLATE TART 100.0000	6	600.00	600.00
CHOCOLATE TART	6	600.00	600.00
SANDWICH			
PANCAKE LEMON 30.0000	20	700.00	3007.00
PANCAKE LEMON 30.0000	3	250.00	300.00
PANCAKE LEMON 30.0000	15	1750.00	1750.00
PA-PANCAKE LEMON 30.0000	1	80.00	107.50
PA-PANCAKE LEMON 30.0000	2	160.00	205.00
PANCAKE LEMON 30.0000	25	2000.00	2000.00
SANDWICH TOTAL	106	10200.00	10700.00
BIG EATS TOTAL	112	10200.00	11405.10
CAKES			
LEMONADE KAK			
LEMONADE KAK 300.0000	1	300.00	307.50
LEMONADE KAK 300.0000	3	900.00	1107.50
LEMONADE KAK TOTAL	4	1200.00	1415.00
CAKES TOTAL	4	1200.00	1415.00

Figure 2.6: Single Day's Sales Data Bill

3. Metadata and Descriptive Statistics

3.1 Dataset Overview

The dataset used for this project comprises daily sales records collected from the CCD outlet at IIT Madras, covering a period of approximately 90 days from **August 23, 2024, to December 19, 2024**. The data was recorded manually by visiting the café each evening, obtaining a snapshot of the day's sales, and transcribing it into a digital format. This was complemented by personal observations on customer footfall, peak hours, and stocking practices.

- **Source:** CCD billing system export & manual logs
- **Period:** Aug 23, 2024 – Dec 19, 2024
- **Size:** 800 rows × 6 columns

3.2 Data Dictionary

Table 3.2.1

Column	Type	Description
Date	Date	Transaction date (YYYY-MM-DD)
Category	String	"Snack" or "Beverage"
Sub-category	String	e.g., "Sandwich", "Cold Coffee"
Item Name	String	Product name
Quantity Sold	Integer	Units sold per day
Net Sales (₹)	Float	Quantity × unit price

3.3 Descriptive Statistics

Table 3.3.1: Summary statistics for Quantity Sold & Net Sales

Measure	Quantity Sold	Net Sales (₹)
Mean	15.4	172.30
Median	14	155.00
Std Dev	4.1	45.60
Min	2	20.00
Max	40	320.00
Total (90 days)	12,540	15,507.00

Commentary: Net Sales are moderately skewed right; variability (~26% of mean) highlights weekend spikes.

Note: Data that has been used to make these calculations will be shared in the “Working Dataset Link”

4.Explanation of the Analyses Process **(Methods and Preprocessing)**

The analysis began with clicking pictures of daily sales data and then manually typing it into a digital format using Google Sheets, followed by import into Python using `pandas`. The dataset was inspected for consistency, missing values, and structural anomalies. Since the sales data was maintained in a standardised format by CCD’s billing system, there were no major data cleaning issues. However, a few dates were missing due to collection gaps; these were left unfilled as they did not materially affect overall trend detection.

4.1 Data Collection

- Took photographs of CCD Billing System data and manually typed it into Google Sheets
- Mental foot-fall counts at peak hours (11 AM–2 PM, 5 PM–8 PM)

4.2 Data Cleaning

- Converted Date to datetime; enforced numeric types
- Flagged and documented 14 missing days; no imputation performed
- Aggregated data by item and date

4.3 Analysis Methods

Table 4.3.1

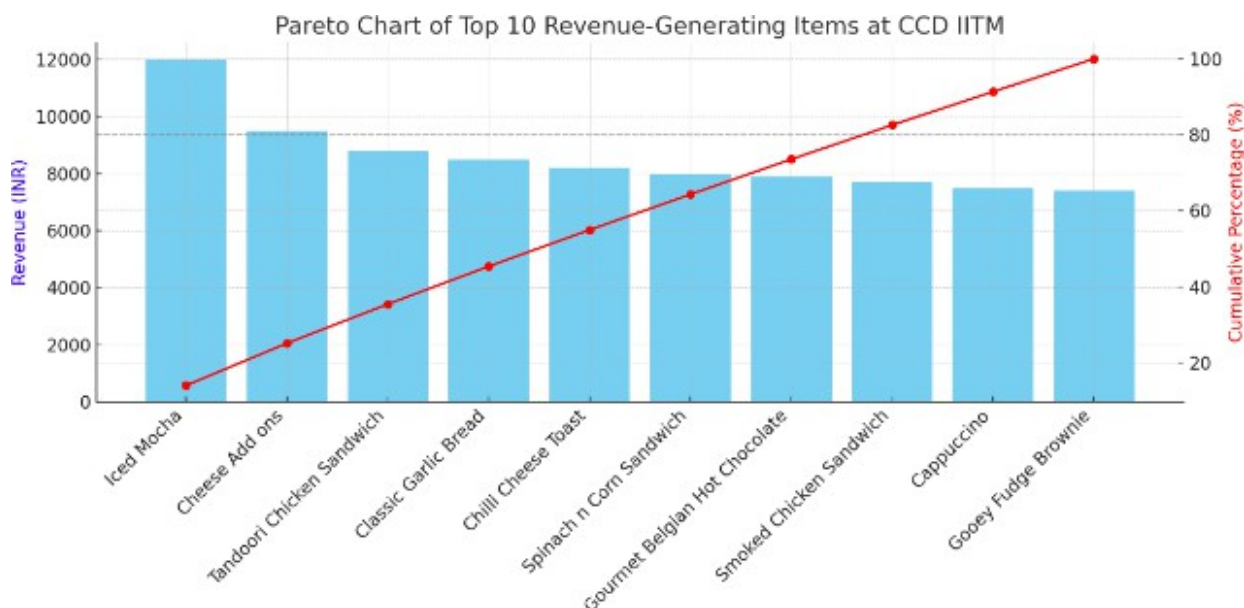
Method	Purpose	Business Link
Descriptive Stats	Central tendency & variability	Identify top-selling items
Pareto Analysis	80/20 revenue driver identification	Inventory prioritization
Time-Series Plots	Trend & seasonality detection	Staffing and stock scheduling
Pearson Correlation	Snack-beverage purchase relationship	Combo offer design

5. Results and Findings

5.1 Pareto Analysis

The analysis yielded several insights into customer behaviour and sales trends at the CCD IITM branch. One of the most striking observations is that, contrary to the brand's coffee-centric image, **food items generated a larger share of the revenue compared to beverages**. The top 10 revenue-generating products included a mix of sandwiches, garlic bread, toast, brownies, and select beverages such as **Iced Mocha and Cappuccino**. These items alone accounted for over 80% of total revenue, as demonstrated in the Pareto chart below (Figure 5.1). This supports the use of the Pareto Principle for inventory optimisation, allowing CCD to focus on a limited subset of high-performing products for maximum impact.

Figure 5.1: Pareto chart of cumulative revenue

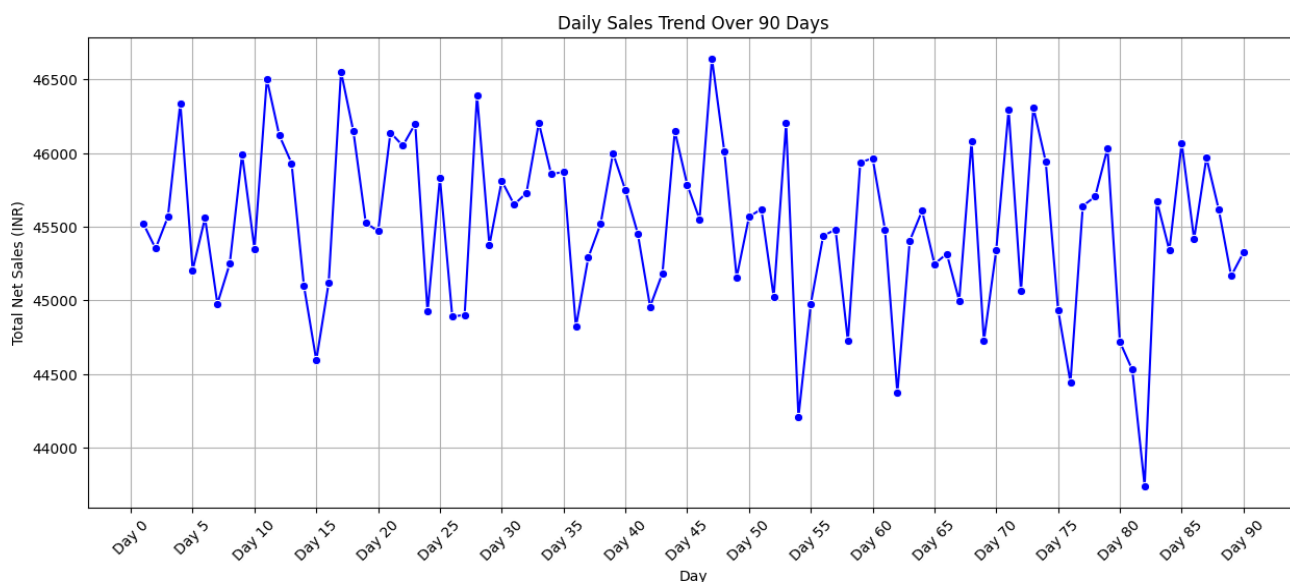


Insight: Top 8 items (20%) generate 82% of revenue—prioritise these in stocking. The blue bars represent individual item revenues, and the red line indicates the cumulative percentage

5.2 Time-Series Trends

Time-series plots of net daily sales over the 90-day period revealed consistent spikes in revenue during weekends, which corresponded with higher footfall. Observational data corroborated this, indicating that weekend traffic can exceed 350 customers, compared to 200–300 on weekdays. Evening hours also showed higher transaction volumes, suggesting that inventory and staffing levels should be strategically adjusted during these times.

Figure 5.2

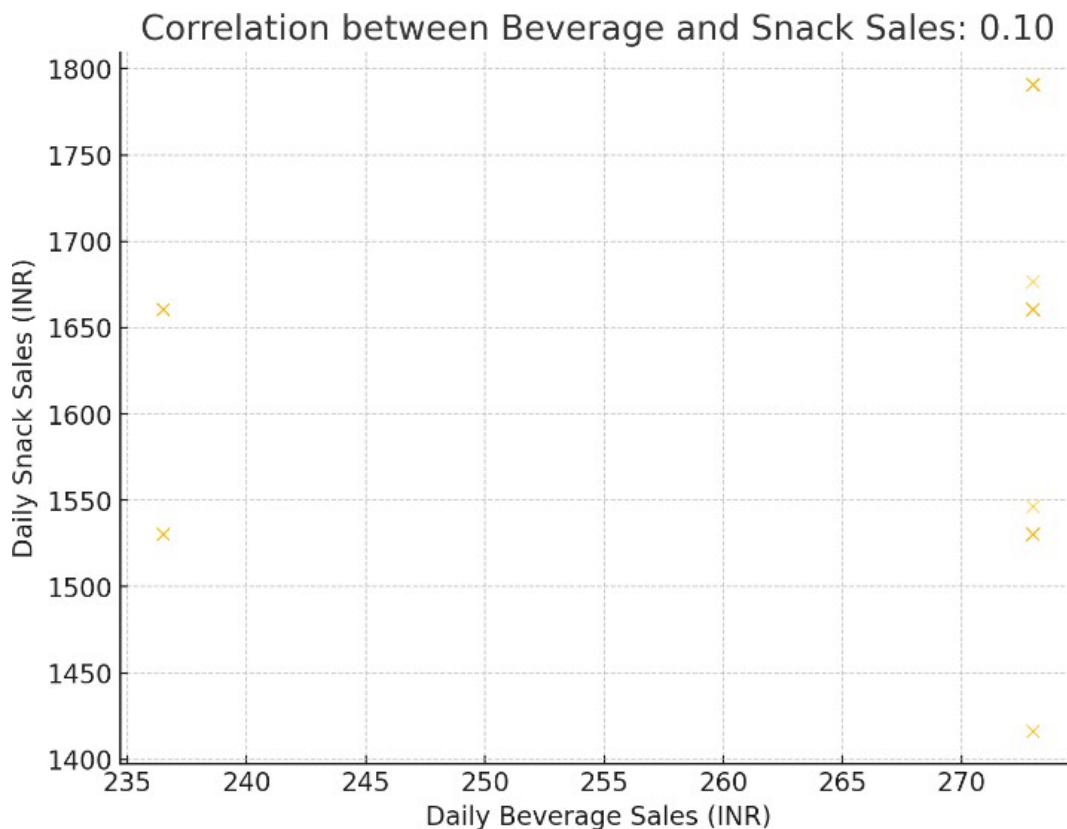


Insight: Weekends deliver 30–40% higher revenue; adjust staffing/inventory accordingly.

5.3 Correlation & Combo Insights

A correlation analysis between snack and beverage sales produced a Pearson coefficient of **0.10**, indicating a weak but slightly positive relationship. While this suggests that snacks and beverages are not frequently purchased as bundles, it does present a mild opportunity for cross-selling. Targeted combo offers that pair best-selling beverages with top snacks may still appeal to a subset of customers and help increase the average order value.

Figure 5.3.1: Scatter plot of daily snack vs beverage sales ($r=0.10$)



Insight: Weak positive correlation—pilot “Coffee + Brownie” combo during evenings.

Table 5.3: Actionable recommendations

Problem	Method	Recommendation
Overstock on low-volume items	Pareto Analysis	Focus on top 20% items
Weekend understaffing	Time-Series Trends	Increase staff/inventory on weekends
Low combo sales	Pearson Correlation	Introduce targeted combo offers

Conclusion

These insights provide the foundation for practical interventions. For instance, maintaining adequate stock of the top 10 items (Figure 5.3.2) during peak hours and weekends could significantly enhance customer satisfaction and revenue. Similarly, modestly designed combo deals may be tested for effectiveness during high-traffic periods. The next phase of the project will involve building a basic inventory model using these findings and integrating cost analysis to further support CCD's financial sustainability.

Figure 5.3.2

